Exp No: 3	Experiments on CISCO PACKET TRACER - Hub and Switch

AIM:

To study the installation of the Cisco Packet Tracer tool and its user interface, and to simulate a network using a hub and a network using a switch.

ALGORITHM / PROCEDURE:

1. Open Cisco Packet Tracer and study the workspace: the device selection box (bottom left), the logical workspace (centre) and the Realtime / Simulation switch (bottom right).
2. Select **End Devices** and place four PCs and a **Hub** on the workspace. Connect each PC to the hub with copper straight-through cables.
3. Place four more PCs and a **Switch** on the workspace and connect them with copper straight-through cables in the same way.
4. Click each PC, go to **Desktop > IP Configuration**, select **Static** and give an IP address (for example PC0 = 10.1.1.1) with subnet mask 255.0.0.0.
5. Switch to **Simulation** mode, click **Add Simple PDU** and send a packet from one PC to another in the hub network. Press **Capture / Forward** and watch how the packet moves.
6. Repeat step 5 in the switch network and compare the behaviour of the hub and the switch in the event list.

DIAGRAM:

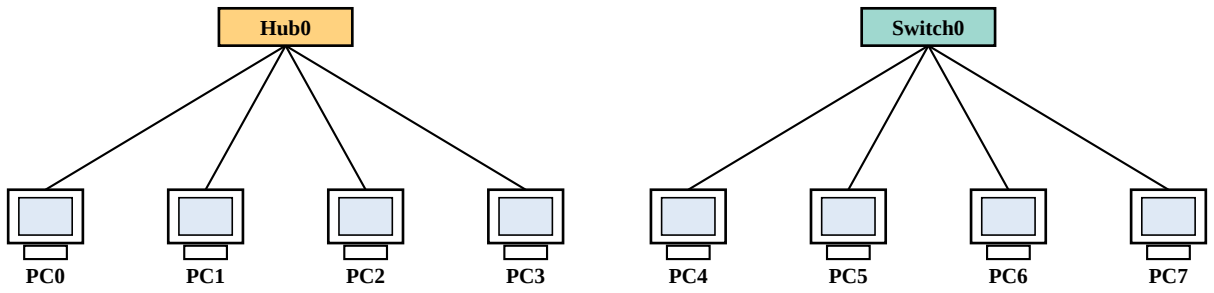
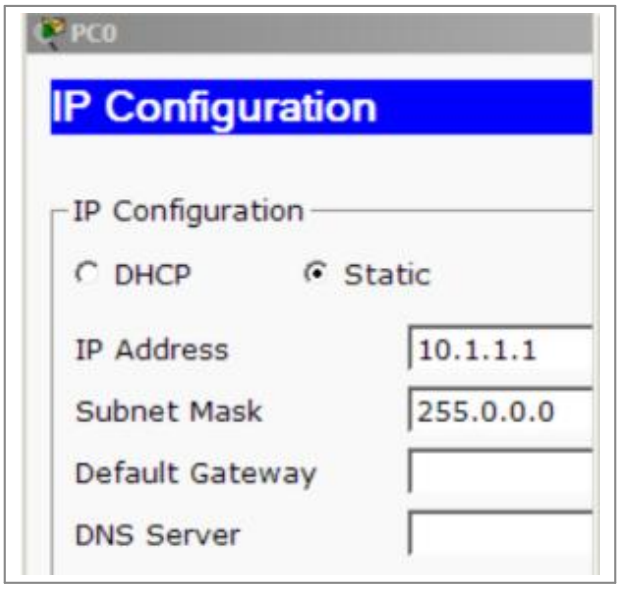
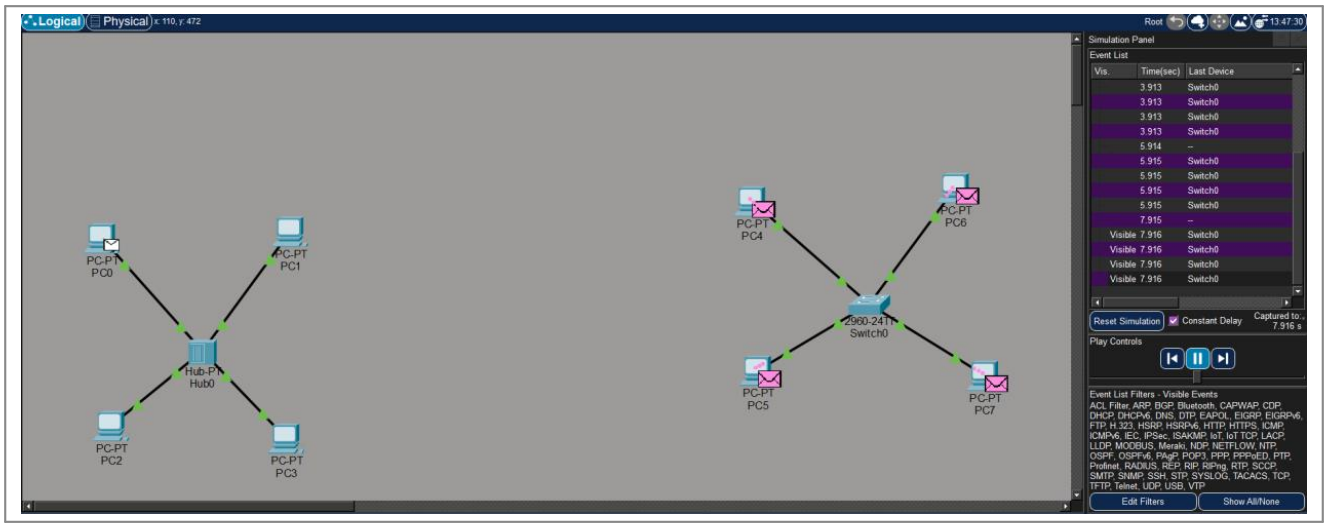


Fig 1: Hub network (left) and switch network (right)

OUTPUT:



IP configuration of PC0 (static address)



Simulation mode: packet transmission in the hub network (left) and the switch network (right)

Comparison of hub and switch

Point	Hub	Switch
OSI layer	Physical layer (Layer 1)	Data link layer (Layer 2)
Forwarding	Repeats the signal to all ports	Forwards the frame only to the destination port
Address table	None	MAC address table (learning)
Collision domain	One for all the ports	One for each port

RESULT:

Thus the Cisco Packet Tracer application was explored and simulations were done for a hub network and a switch network.